Magpie Industries SC — Project Plan

> *A Star Citizen community web application for orgs, friend groups, and solo players. Inspired by uexcorp.space, built to go further with org management, group coordination, and mission guides.*

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1. App Hygiene

1.1 Project Name

- **App Name:** Magpie Industries SC
- **Domain:** *[TBD — e.g., magpieindustries.space]*
- **Tagline:** *[TBD — short phrase describing the app's purpose]*

1.2 Target Audience

- [ ] Solo players looking for tools and mission guides
- [ ] Friend groups coordinating play sessions
- [ ] Orgs managing members, roles, and operations
- *[Add more as needed]*

1.3 Branding & Identity

- **Logo:** *[TBD]*
- **Color Palette:** *[TBD — dark mode primary? accent colors?]*
- **Typography:** *[TBD]*
- **Tone of Voice:** [ ] In-universe lore-friendly

1.4 Accessibility & Standards

- [ ] WCAG 2.1 AA compliance target
- [ ] Mobile-first responsive design
- [ ] Keyboard navigation support
- [ ] Screen reader friendly
- *[Add or remove as needed]*

1.5 Legal / Compliance

- **Privacy Policy:** *[TBD]*
- **Terms of Service:** *[TBD]*
- **Data Retention Policy:** *[TBD]*
- **Star Citizen IP/Fan Site Guidelines:** *[TBD — review CIG's fan content policy]*

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2. Architecture

2.1 Tech Stack

Layer	Technology	Notes
Frontend	React	*[SSR? SPA? SSG?]*
Backend	Node	
Database	PostgreSQL	
Auth	OAuth	
Hosting	Render	

CDN	*[TBD — Cloudflare, etc.]*	
CI/CD	GitHub Actions	

2.2 Data Sources

Primary: Game file datamining

- SC stores game data in `.p4k` archives in the install directory
- Tools like **sctool** or **StarFab** unpack these into readable XML/JSON
- Data lives under `Data/Libs/Foundry/Records/` once extracted
- Covers: commodity prices, ore compositions, refinery yields/times, ship specs, weapon/component stats, item properties, location data
- Caveat: must re-extract after each patch; some values are server-side only (dynamic pricing, live inventory)

Secondary: Existing community APIs & datasets

- [ ] **scunpacked** — pre-parsed game data (good starting point to reduce maintenance)
- [ ] **UEX API** (if public / with permission)
- [ ] **starcitizen.tools** wiki data
- [ ] **sc-trade.tools** — trade/commodity data

Tertiary: Community-contributed

- [ ] User-submitted tips, strategies, and corrections
- [ ] Community-reported dynamic prices and availability

Strategy: Start with existing community-parsed data (less maintenance). Build a custom `.p4k` extraction pipeline later for data nobody else provides. Layer community contributions on top for real-time/dynamic info.

> ***Note:*** *CIG generally tolerates datamining for community tools, but review their current ToS before publishing.*

2.3 Authentication & Roles

Role	Description
Visitor	Browse public tools, guides, and info
Registered User	Save preferences, join groups, bookmark guides
Group Member	Access shared group dashboards and planning
Org Member	Access org tools, rosters, event scheduling
Org Leader	Manage org settings, members, roles
Admin	Full site administration
[Add more]	

2.4 High-Level Architecture Diagram

```
_[TBD — draw or describe the system layout]_

Example placeholder:

  [Client SPA] <---> [API Gateway / Backend]
                          |         |
                     [Database]  [External APIs]
```

```
                    |
                [Cache Layer]
```

2.5 Key Architecture Decisions

- *[TBD — e.g., monolith vs microservices]*
- *[TBD — real-time features via WebSockets or polling?]*
- *[TBD — image/file storage strategy]*
- *[TBD — caching strategy]*

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3. Key Features

3.1 Community & Social

Feature	Priority	Status	Notes
User profiles	*[H/M/L]*	*[TBD]*	RSI handle linking, avatar, bio
Friend lists	*[H/M/L]*	*[TBD]*	
Group creation	*[H/M/L]*	*[TBD]*	Casual friend groups, not full orgs
Org pages	*[H/M/L]*	*[TBD]*	Public-facing org profile & roster
Org management dashboard	*[H/M/L]*	*[TBD]*	Member roles, permissions, settings
Event scheduling	*[H/M/L]*	*[TBD]*	For orgs and friend groups
In-app messaging / chat	*[H/M/L]*	*[TBD]*	
Activity feed	*[H/M/L]*	*[TBD]*	
[Add more]			

3.2 Tools (Inspired by UEX)

Feature	Priority	Status	Notes
Trade route planner	*[H/M/L]*	*[TBD]*	Commodity prices, profit calc
Ship/vehicle database	*[H/M/L]*	*[TBD]*	Specs, pricing, comparisons
Loadout planner	*[H/M/L]*	*[TBD]*	Ship component fitting
Mining calculator	*[H/M/L]*	*[TBD]*	Input ore type, rock composition %; outputs refined value & profit per SCU
Salvage calculator	*[H/M/L]*	*[TBD]*	RMC values, material yields, profit estimates
Refinery optimizer	*[H/M/L]*	*[TBD]*	Compare methods (Dinyx Solventation, Cormack, Ferron Exchange, etc.) by time, yield, and profit
Profit simulator	*[H/M/L]*	*[TBD]*	Investment vs income, ROI tracking
[Add more]			

3.3 Wikelo Mission Guides

> ***What are Wikelo missions?*** *Wikelo is a Banu trader in the Stanton system — the first non-human NPC in Star Citizen. He operates collection contracts where players gather items and trade them for rewards (ships, weapons, armor, components). Contracts are accessed at*

> *three Wikelo Emporium stations:* ***Dasi Station*** *(Hurston),* ***Selo Station*** *(Yela), and* ***Kinga Station*** *(microTech).*
>
> ***Reputation tiers:*** *New Customer → Very Good Customer → Very Best Customer. Higher ranks unlock better contracts and exclusive rewards.*

Feature	Priority	Status	Notes
Contract catalog	*[H/M/L]*	*[TBD]*	Full list of Wikelo contracts with requirements & rewards
Reputation progression guide	*[H/M/L]*	*[TBD]*	How to rank up through the 3 tiers efficiently
Item gathering walkthroughs	*[H/M/L]*	*[TBD]*	Where to find each required item (fauna, ore, scrip, etc.)
Emporium location guides	*[H/M/L]*	*[TBD]*	Maps/directions to Dasi, Selo, Kinga stations
Reward database	*[H/M/L]*	*[TBD]*	Ships, weapons, armor, components — with specs
Favor economy breakdown	*[H/M/L]*	*[TBD]*	How Wikelo Favors work as currency, best conversion paths
Recommended loadouts	*[H/M/L]*	*[TBD]*	Ships & gear for gathering (Ursa Rover, Prospector, etc.)
Seasonal/rotating contracts	*[H/M/L]*	*[TBD]*	Track which contracts are active vs inactive
Tips & strategies	*[H/M/L]*	*[TBD]*	Optimal routes, time-saving tricks
Community comments / tips	*[H/M/L]*	*[TBD]*	User-submitted additions
[Add more]			

3.4 Org & Group Tools

Feature	Priority	Status	Notes
Shared fleet tracker	*[H/M/L]*	*[TBD]*	Who owns what ships
Crew assignment board	*[H/M/L]*	*[TBD]*	Assign members to ship crew positions per operation (see below)
Operation planner	*[H/M/L]*	*[TBD]*	Multi-crew mission coordination, ties into crew assignments
Role assignment system	*[H/M/L]*	*[TBD]*	Custom org ranks & permissions
Org treasury / fund tracker	*[H/M/L]*	*[TBD]*	
Recruitment board	*[H/M/L]*	*[TBD]*	Orgs looking for members
[Add more]			

> ***Crew Assignment System — Detail***
>
> *Orgs list their available ships for an operation, then assign members to crew positions. Each ship gets its own crew roster. Members see their assignment, ship, and role at a glance. Org leaders drag-and-drop or assign members to open slots.*
>
> *Operations are built from* ***operation templates*** *based on type. Each template defines relevant roles. Ships can be mixed (industrial + security escort, etc.).*
>
> ---
>
> ***General Ship Roles*** *(available across all operation types)*
>
> | *Position* | *Description* |
> |---|---|
> | *Captain / Pilot* | *Commands the ship, handles flight* |
> | *Co-Pilot* | *Assists with navigation and flight ops* |

Flight Operations	*Coordinates fleet movement and comms*
Engineer	*Manages power, shields, and repairs*
Gunner	*Operates turrets and weapon systems*
Hangar Services	*Pre-flight loadout, refuel, rearm, repair*
Medical	*Crew recovery and support*

Mining Operation Roles *(single ship or multi-ship fleet)*

Position	*Description*
Mining Lead	*Coordinates scan targets and extraction priority*
Laser Operator	*Operates mining lasers on multi-crew ships (Mole)*
Scan Operator	*Runs prospecting scans, identifies valuable rocks*
Refinery Coordinator	*Manages refinery jobs and method selection*
Hauler / Transport	*Ferries refined ore from refinery to sell point*

Salvage Operation Roles

Position	*Description*
Salvage Lead	*Coordinates salvage targets and priorities*
Salvage Operator	*Operates salvage beams (Reclaimer, Vulture)*
Hull Scraper	*Manual FPS salvage / EVA hull stripping*
Material Transport	*Hauls salvaged materials to sell point*

Cargo / Hauling Operation Roles

Position	*Description*
Cargo Lead	*Plans route, manages buy/sell decisions*
Loadmaster	*Oversees cargo loading and manifest*
Cargo Handler	*Loads/unloads freight at terminals*
Freighter Pilot	*Flies the cargo ship*

Security / Escort Roles *(attachable to any operation type)*

Position	*Description*
Security Lead	*Coordinates escort fleet and threat response*
Escort Pilot	*Flies dedicated escort fighter/gunship*
Ship Gunner	*Turret operator on industrial or escort ships*
Ground Security	*FPS protection at landing zones or boarding defense*

Overwatch / Scout	*Advance recon, monitors for incoming threats*

Combat Operation Roles

Position	*Description*
Fleet Commander	*Overall tactical command of the operation*
Wing Lead	*Commands a sub-group of fighters*
Bomber Pilot	*Flies torpedo/bomber runs on capital targets*
Landing Crew	*Ground team for boarding, FPS, or objective ops*
SAR / Medevac	*Search and rescue, downed pilot recovery*

Roles are mix-and-match. An org can combine mining + security for a protected mining op, or cargo + escort for a guarded hauling run. Templates are customizable per org.

3.5 Resources & Reference

Feature	Priority	Status	Notes
Locations database	*[H/M/L]*	*[TBD]*	Planets, stations, POIs
Commodity price index	*[H/M/L]*	*[TBD]*	
Patch notes / changelog	*[H/M/L]*	*[TBD]*	Track SC updates
Beginner's guide	*[H/M/L]*	*[TBD]*	
FAQ / known workarounds	*[H/M/L]*	*[TBD]*	
[Add more]			

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4. Milestones & Phases

Phase 1 — MVP (Core Tools & Auth)

Get the foundation live: auth, basic profiles, and the most useful standalone tools.

- [ ] Project scaffolding (React, Node, PostgreSQL, OAuth)
- [ ] User authentication & basic profiles (RSI handle, avatar, bio)
- [ ] Mining calculator (ore composition input, profit per SCU output)
- [ ] Salvage calculator (RMC values, material yields)
- [ ] Refinery optimizer (compare methods by time, yield, profit)
- [ ] Ship/vehicle database (specs, pricing, comparisons)
- [ ] Responsive dark-mode UI with lore-friendly branding

Phase 2 — Guides & Content

Add the knowledge layer — Wikelo guides, trade tools, and reference data.

- [ ] Wikelo contract catalog (full list with requirements & rewards)
- [ ] Wikelo reputation progression guide
- [ ] Item gathering walkthroughs (locations for fauna, ore, scrip)

- [ ] Emporium location guides (Dasi, Selo, Kinga)
- [ ] Reward database (ships, weapons, armor with specs)
- [ ] Favor economy breakdown
- [ ] Trade route planner (commodity prices, profit calc)
- [ ] Locations database (planets, stations, POIs)
- [ ] Beginner's guide

Phase 3 — Community & Social

> *Let users connect — friend groups, activity feeds, and shared content.*

- [ ] Friend lists
- [ ] Group creation (casual friend groups)
- [ ] Event scheduling (for groups)
- [ ] Activity feed
- [ ] Community comments / tips on guides
- [ ] Loadout planner (ship component fitting)
- [ ] Profit simulator (investment vs income, ROI)

Phase 4 — Org Tools

> *Full org support — management dashboards, fleet tracking, and recruitment.*

- [ ] Org pages (public/private toggle, profile & roster)
- [ ] Org management dashboard (member roles, permissions, settings)
- [ ] Shared fleet tracker (who owns what ships)
- [ ] Crew assignment board (assign members to ship positions per operation)
- [ ] Operation planner (multi-crew mission coordination, tied to crew assignments)
- [ ] Role assignment system (custom org ranks & permissions)
- [ ] Org treasury / fund tracker
- [ ] Recruitment board
- [ ] Event scheduling (org-level)

Phase 5 — Real-Time & Scale

> *Add live features, polish performance, and harden for growth.*

- [ ] Real-time features (scope TBD — chat, notifications, live tracking)
- [ ] In-app messaging / chat
- [ ] Seasonal/rotating Wikelo contract tracker
- [ ] CDN setup & caching optimization
- [ ] Performance audits & load testing
- [ ] Patch notes / changelog tracking for SC updates
- [ ] FAQ / known workarounds

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5. Open Questions

> *Things we still need to decide or research.*

- [x] ~~What are Wikelo missions?~~ — Resolved. Banu trader collection contracts. See Section 3.3.
- [x] ~~Should orgs be able to set their pages to private or public?~~ — Yes, orgs can toggle visibility.

- [ ] **Real-time features** — Exploring options (chat, live fleet tracking, notifications, etc.). Need to decide scope and tech (WebSockets, SSE, polling).
- [x] ~~Data sourcing strategy~~ — Resolved. Game file datamining + community APIs + user contributions. See Section 2.2.
- [ ] *[Monetization strategy? (free, donations, premium tiers)]*
- [ ] *[Add more]*

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